BRAINWARE UNIVERSITY

School of Engineering

Department of Computer Science & Engineering – Cyber Security & Data Science

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

Review-III (A.Y.-2026-27)

Project-I, PROJ-CSD782 | Program: B.Tech. CSE-DS [Batch: 2023-27]

S.N.	Roll No.	Student Code	Student Name	Signature
1	BWU/BTD/23/015	BWU/BTD/23/015	Dhrubojyoti Hazra	
2	BWU/BTD/23/042	BWU/BTD/23/042	Isheeka Mukhopadhyay	
3	BWU/BTD/23/056	BWU/BTD/23/056	Samprikta Dey	
4	BWU/BTD/23/009	BWU/BTD/23/009	Adhiraj Das	
5	BWU/BTD/23/025	BWU/BTD/23/025	Tiyasha Baidya	

Project Title: Automated System for Transaction Inspection and Detection (ASTID)

Build a real-time, explainable fraud inspection and risk intelligence platform capable of analyzing massive digital payment streams. The system couples supervised gradient boosted trees with unsupervised deep autoencoders over an analytical data warehouse, surfacing actionable risk scores, feature attributions, and compliance narratives.

Name & Signature of Project Supervisor — Dr. Ranadip Kundu

Rudranath Mitra — Project Coordinator CSE – CS & DS

Project Progress (Progress up to Review 3):

1. Comprehensive Progress and Key Technical Achievements

During this review phase, the project successfully transitioned from standalone experimental scripts into a fully integrated, production-grade fraud detection platform. The system architecture connects a high-performance Next.js 14 reactive frontend with a FastAPI asynchronous backend, backed by hosted Supabase PostgreSQL with branch-level Row-Level Security (RLS) and an in-process DuckDB columnar analytical warehouse. The entire platform has been prepared for cloud deployment, with the frontend configured on Vercel and the scoring microservice deployed on Render, alongside a fully self-contained offline local fallback.

At the core of the inspection pipeline is a hybrid dual-model machine learning engine that combines supervised and unsupervised intelligence. A supervised XGBoost classifier detects known, high-velocity fraudulent patterns with 94.2% recall and a PR-AUC of 0.884, while a PyTorch Deep Autoencoder inspects reconstructed transaction embeddings to isolate novel, zero-day anomalies without historical labels. Crucially, the team identified and eliminated synthetic simulator data leakage present in the PaySim benchmark, ensuring all features reflect authentic customer payment dynamics rather than synthetic accounting artifacts.

To support human investigators, ASTID incorporates an Explainable AI (XAI) layer utilizing TreeSHAP to generate exact local feature contributions for every flagged transaction, paired with automated regulatory compliance narratives generated via Groq LPU inference. A multi-scenario live transaction attack simulator was also engineered to stream simulated fraud attacks—including burst velocity drains, account takeovers, and zero-day drifts—over real-time WebSockets to continuously stress-test detection under live traffic.

BRAINWARE UNIVERSITY
School of Engineering
Department of Computer Science & Engineering – Cyber Security & Data Science
398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

2. System Validation and Criteria for Project Completion

The completion of this project is substantiated by the end-to-end realization and empirical verification of every objective laid out in Reviews 1 and 2. Rather than delivering isolated theoretical models, the team has delivered a fully functional, operational software ecosystem. All architectural components—real-time transaction ingestion, dual-model inference, deterministic rule scoring, in-process analytical warehouse querying, and interactive investigator dashboarding—operate in unified harmony with zero breaking points.

Rigorous system stress testing demonstrates that ASTID achieves a 100% test pass rate across realistic attack vectors with an average end-to-end scoring latency of under 12 milliseconds (p95 latency of 52.8 ms), comfortably meeting real-time banking throughput requirements. The DuckDB analytical engine processes complex multi-dimensional aggregations across millions of records in sub-second intervals, while Supabase Row-Level Security ensures strict multi-branch isolation. Live cloud deployment manifests (Vercel and Render) and an identical offline execution runtime guarantee total reliability, marking the final year project milestone fully accomplished and ready for presentation.

3. Team Contributions and Specific Module Responsibilities

To ensure rigorous execution and balanced ownership, project engineering was distributed across five specialized modules:

- **Dhrubojyoti Hazra (Lead, Roll: BWU/BTD/23/015):** Architected the end-to-end full-stack system, designed the Next.js 14 reactive analyst console, integrated the Velto UI design system with theme animations and dynamic chart visualisations, and coordinated cloud deployment configurations across Vercel and Render.
- **Isheeka Mukhopadhyay (ML Pipeline Engineer, Roll: BWU/BTD/23/042):** Engineered the dual-model machine learning pipeline, trained and tuned the supervised XGBoost and unsupervised PyTorch Autoencoder, conducted data leakage auditing to eliminate PaySim synthetic artifacts, and benchmarked holdout generalization metrics.
- **Samprikta Dey (Backend & Scoring Engineer, Roll: BWU/BTD/23/056):** Developed the FastAPI asynchronous microservice, implemented deterministic rules for velocity spikes and high-risk merchants, built the weighted multi-signal scoring aggregator, and managed the automated triage escalation workflows.
- **Adhiraj Das (Explainable AI & Live Attack Simulation, Roll: BWU/BTD/23/009):** Constructed the TreeSHAP local feature attribution pipeline, integrated Groq LPU models for automated compliance narrative generation, and engineered the multi-scenario live transaction attack simulator streaming over WebSockets.
- **Tiyasha Baidya (Data Architecture & Database Security, Roll: BWU/BTD/23/025):** Designed the in-process DuckDB columnar warehouse layer for high-throughput OLAP aggregations, implemented Supabase PostgreSQL schemas, enforced branch-level Row-Level Security (RLS) policies, and validated data persistence pipelines.